Airplane Boarding Strategy

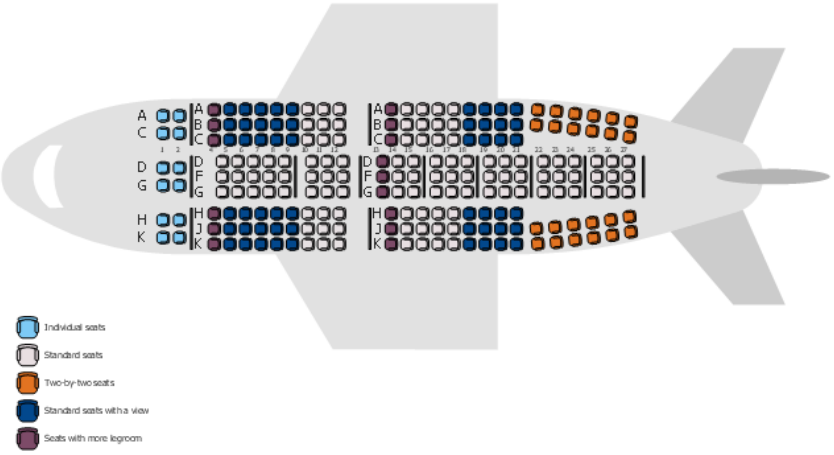


Abbildung 1: Seatplan of a large aircraft https://www.conceptdraw.com.

Motivation

Hardly any other system offers as much potential for improvement as the boarding of an airplane. Hardly anyone has not thought: wouldn't it be smarter if those sitting by the window boarded first? In this project, we want to test this and other questions with the help of a simulation model. We want to analyse the extent to which these naive ideas for improvement hold up not only in theory but in a world that also reflects people's imperfect behaviour.

Model

We tackle the question using a classic queuing model since aisles of a plane do not provide any space for takeovers. In this specification we will only consider an airplane with a single centre aisle and one entry point at the front.

The single entry of the air plane is modelled using a combination of an entity source and and a limitless queue. In the source, passengers are created and assigned a seat, whereas order of the seat assignment reflects the order for boarding. The queue object models the stairs/walkway outside the aircraft in which passengers are required to wait until there is sufficient space to enter the plane.

Moreover, the centre aisle is further divided into sequential single server queues - one queue per seat-row with a maximum capacity of 2 persons each. This reflects that one person is busy entering the row and a second person is able to stand and wait between any two rows. When a person is ready for service, the following distinction is made:

- In case the person does not require one of the seats in the row, the service time is zero and the passenger can proceed to the next queue in the sequence if any only if the queue is not full.
- In case the person requires one of the seats in the row, the service time will depend on, whether the seat(s) between the designated seat and the aisle are already taken.
- For the latter, use and define a parameter-map between all possible configurations (required seat and which seats are already taken) and the corresponding average service time (i.e. time to manage luggage plus time to get to and take the seat).

To Do's

Task 1

The stated model definition is, on purpose, quite loose and leaves space for fine tuning and specification. Think about the model in detail and specify the model in more detail (e.g. which service time distribution

is used, do all passengers arrive at once, how many rows and seats does your airplane have, ...). Use the event graph logic to sketch your model.

Task 2

Implement your model using a random seating strategy (i.e. the seats are assigned randomly). Perform face validation and experiment with different parameter values (times to take the seat). Fix one parameter setting for Tasks 4 and 5.

Task 3

Implement and experiment with different boarding strategies. Measure the potential time advantage of the strategies.

Task 4

Implement, that human beings are not always compliant. I.e. consider a chance, that individuals do not board in the sequence they are asked to. Measure the time and contrast them to the perfect world scenarios.

Task 5

Extend your model in at least one of the following ways:

- Make it more realistic,
- Model a plane which can be boarded from the front and the back, or
- Model a plane which has more then one centre aisle.